1 DECEMBRIE 1918 UNIVERSITY OF ALBA IULIA
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

TIERED CLASSIFICATION AND CONFIDENCE-BASED ROUTING FOR CHEST RADIOGRAPHS

BACHELOR'S THESIS

Alperen Yörük

Thesis Advisor:
Asist. univ. drd. Daniela-Maria Cristea

Alba Iulia
2026

Abstract

This thesis proposes a tiered classification and confidence-based routing system to distinguish pneumothorax from normal findings in chest radiographs, addressing two structural weaknesses of conventional single-stage classifiers: their uniform computational cost and their inability to report calibrated uncertainty or to abstain. A lightweight MobileNetV2 screening stage (Tier 1) resolves confident cases directly and escalates only low-confidence images to a heavier EfficientNet-B4 or Ark+ Swin stage (Tier 2). The escalation threshold adapts online to the stream of recent confidences, Monte Carlo Dropout and test-time augmentation estimate predictive uncertainty, temperature scaling calibrates the probabilities, and conformal prediction equips every decision with a finite-sample coverage guarantee. A Stable-Diffusion pipeline augments the under-represented pneumothorax class. The method was evaluated with a genuine fifteen-configuration ablation study on the NIH ChestX-Ray14 dataset and a zero-shot cross-dataset evaluation on CheXpert. The lightweight tier resolves roughly three quarters of all cases on its own; the uncertainty stack reduces the Expected Calibration Error from 0.074 to 0.050 without harming discrimination; and the full system, with an Ark+ Swin backbone and mixup/CutMix regularisation, attains the best overall balance, with an area under the ROC curve of 0.924 and a conformal predictor that meets its 95% coverage target. The system is delivered as a complete, reproducible application with a continuous-integration pipeline that keeps every reported number regenerable from the evaluation artifacts.

Keywords: Chest Radiography, Pneumothorax Classification, Tiered Classification, Conformal Prediction, Monte Carlo Dropout.

Contents

List of Figures

List of Tables

Chapter 1

Introduction

1.1 Context and Motivation

Chest radiography is the most frequently performed diagnostic imaging examination in the world. It is inexpensive, fast, low in radiation dose, and almost universally available, which makes it the first-line investigation for a wide range of thoracic conditions. Among these conditions, pneumothorax — the presence of air in the pleural space, which causes the lung to collapse partially or completely — is of particular importance because a large or tension pneumothorax is a medical emergency that requires immediate intervention. A delayed or missed diagnosis can lead to rapid clinical deterioration and, in the worst case, death.

The interpretation of chest radiographs, however, remains a demanding task. The visual signs of an early pneumothorax can be subtle, the volume of examinations a radiologist must read is high, and fatigue is a well-documented source of diagnostic error. These pressures have motivated a sustained research effort into computer-aided diagnosis, and over the last decade deep convolutional neural networks have come to dominate the field, reaching expert-level performance on several chest pathologies [1,2].

Despite this progress, the deep-learning systems that are deployed in practice share two structural weaknesses. First, they are typically *monolithic*: a single, often large, network is applied to every image regardless of how easy or hard the case is, so the most computationally expensive model is paid for even on the overwhelming majority of clearly normal studies. Second, and more critically for clinical use, they are usually *over-confident and silent about their own uncertainty*: a standard classifier is forced to emit a definitive label for every input and provides no calibrated, statistically meaningful statement about how much that label can be trusted. In a safety-critical setting, a model that cannot say "I am not sure, please have a human look at this" is difficult to integrate responsibly.

1.2 Problem Statement

This thesis addresses both weaknesses simultaneously. The central question is whether a chest-radiograph classifier can be made *both* more computationally efficient *and* more trustworthy, without sacrificing diagnostic performance. Concretely, we target three deficiencies of the conventional single-stage approach:

1. **Uniform computational cost.** Every image is processed by the same heavy

model, even though most cases are easy. This wastes computation, energy, and time.

2. **Absence of calibrated uncertainty.** The raw softmax output of a deep network is not a reliable probability, and a single point prediction carries no guarantee about the likelihood that it is correct.

3. **Forced decisions.** The model must commit to a class for every input and has no principled mechanism to abstain or to defer an ambiguous case to a human expert.

1.3 Proposed Approach

We propose a two-stage *tiered* architecture coupled with confidence-based routing. The first stage (Tier 1) is the lightweight and fast MobileNetV2 model [3], which acts as a screening filter. When Tier 1 is confident, its decision is accepted directly. When it is not, the image is escalated to a second, more capable stage (Tier 2), built on EfficientNet-B4 [4] or on the Ark+ foundation model with a Swin Transformer backbone [5, 6]. Because the lightweight stage resolves the large majority of cases, the average computational cost is reduced substantially while the difficult cases still receive the full power of the deep model.

On top of this cascade, the system quantifies and acts on uncertainty in three complementary ways. Monte Carlo Dropout [7] and test-time augmentation provide an estimate of the model's epistemic uncertainty, the escalation threshold itself adapts online to the stream of recent confidences, and the Conformal Prediction framework [8, 9] converts the model's scores into prediction *sets* that are guaranteed to contain the true label with a user-chosen probability. Finally, because the minority (pneumothorax) class is under-represented in the training data, a Stable-Diffusion [10] synthetic-augmentation pipeline is used to enrich it, and HiResCAM [11] saliency maps are produced to make the system's decisions interpretable.

1.4 Contributions

The principal contributions of this thesis are the following:

- A complete, reproducible tiered classification and confidence-based routing system for pneumothorax detection, integrating MobileNetV2, EfficientNet-B4, and an Ark+ Swin backbone behind a single adaptive router.

- The integration of three uncertainty mechanisms — Monte Carlo Dropout, test-time augmentation, and conformal prediction — into the routing decision, together with temperature-scaling calibration.

- A genuine, fully evaluated ablation study of fifteen system configurations (A1–A15) on the NIH ChestX-Ray14 test set, isolating the contribution of each component, plus a zero-shot cross-dataset evaluation on CheXpert.

- An end-to-end software system — a FastAPI back end, a React front end, Docker packaging, MLflow experiment tracking, and a continuous-integration pipeline — that demonstrates the method in a deployable form.

1.5 Structure of the Thesis

The remainder of this thesis is organised as follows. Chapter 2 reviews the relevant literature on deep learning for chest radiography, efficient and cascaded architectures, uncertainty estimation, conformal prediction, synthetic data, and explainability. Chapter 3 presents the proposed methodology in detail, including the tiered architecture, the routing logic, the uncertainty and conformal components, and the synthetic-augmentation pipeline. Chapter 4 describes the datasets, the training configuration, the evaluation protocol, and the design of the ablation study. Chapter 5 reports the quantitative results. Chapter 6 interprets these findings, discusses their clinical implications, and states the limitations of the work. Chapter 7 concludes and outlines directions for future research.

Chapter 2

Related Work

This chapter situates the thesis within the existing literature. We review, in turn, deep learning for chest radiography and the public benchmarks that enabled it, efficient and cascaded architectures, foundation models for medical imaging, uncertainty estimation and calibration, conformal prediction, synthetic data and augmentation, and explainability. The chapter closes by positioning the proposed system relative to this body of work.

2.1 Deep Learning in Chest Radiography

The modern era of medical image classification began with the success of deep convolutional neural networks on natural images, starting with AlexNet [12] and continuing through VGG [13], residual networks [14], and densely connected networks [15]. The availability of very large labelled chest-radiograph datasets made it possible to transfer these architectures to the medical domain. The NIH ChestX-Ray14 dataset [16], with over one hundred thousand frontal radiographs labelled for fourteen pathologies, and the Stanford CheXpert dataset [2], with its explicit treatment of label uncertainty, became the standard benchmarks; MIMIC-CXR [17] later added free-text reports at a similar scale.

Building on these resources, CheXNet [1] demonstrated radiologist-level pneumonia detection with a 121-layer DenseNet, and a large number of follow-up studies refined the architecture, the loss function, and the labelling strategy. Almost all of these systems share two characteristics that this thesis sets out to change: they are single-stage, applying one fixed network to every image, and they report only a point prediction, with no calibrated uncertainty and no mechanism to abstain. Broad surveys of deep learning in medical image analysis [18] and landmark results in other modalities, such as dermatologist-level skin-cancer classification [19], confirm both the promise of the approach and the recurring concern that such models are poorly calibrated and difficult to trust in isolation.

2.2 Efficient and Cascaded Architectures

A parallel line of research has sought to reduce the computational cost of deep networks. MobileNets [3, 20] use depthwise-separable convolutions and inverted residual blocks to achieve a strong accuracy-to-cost ratio on mobile and embedded hardware, while

EfficientNet [4] introduces compound scaling to balance network depth, width, and input resolution. More recently, vision transformers [21] and their hierarchical variant, the Swin Transformer [6], have matched or exceeded convolutional networks at the cost of larger models and more data.

The idea of routing inputs through models of increasing capacity — variously called cascades, early-exit networks, or conditional computation — predates deep learning but fits it naturally: most inputs are easy and can be resolved cheaply, so spending the full model only on the hard cases reduces the average cost. The present work adopts exactly this principle, pairing a MobileNetV2 screening stage with a heavier EfficientNet-B4 or Swin-based stage, and adds a confidence-based router that decides, per image, whether escalation is warranted.

2.3 Foundation Models for Medical Imaging

Transfer learning from ImageNet [22, 23] has long been the default initialisation for medical image classifiers, but the domain gap between natural and medical images limits its benefit. This has motivated medical-specific pretraining. The Ark family of models [5] accrues and reuses knowledge across many public chest-radiograph datasets and their heterogeneous label sets, producing a backbone that transfers more robustly to downstream chest tasks than a generically pretrained network. In this thesis, the Ark+ model with a Swin Transformer backbone is used as one of the two Tier-2 options, and is compared head-to-head with a conventionally pretrained EfficientNet-B4.

2.4 Uncertainty Estimation and Calibration

Quantifying predictive uncertainty is essential for safe deployment. Gal and Ghahramani [7] showed that dropout, when left active at test time, can be interpreted as approximate Bayesian inference: performing T stochastic forward passes (Monte Carlo Dropout) yields a distribution over predictions whose variance estimates epistemic uncertainty. Deep ensembles [24] provide a strong, if expensive, alternative, and Kendall and Gal [25] formalise the distinction between aleatoric and epistemic uncertainty in deep vision models. Test-time augmentation, which aggregates predictions over augmented copies of an input, is a complementary technique that also exposes the sensitivity of a prediction to plausible input perturbations [26].

A related but distinct problem is calibration: whether a model's stated confidence matches its empirical accuracy. Modern deep networks are known to be poorly calibrated [27], typically over-confident, and temperature scaling is a simple and effective post-hoc remedy. Calibration quality is commonly summarised by the Expected Calibration Error [28]. This thesis applies Monte Carlo Dropout and test-time augmentation for uncertainty, temperature scaling for calibration, and reports the Expected Calibration Error throughout.

2.5 Conformal Prediction

Conformal Prediction [8, 9] is a framework that turns any point predictor into a set predictor with a finite-sample, distribution-free coverage guarantee: for a chosen error rate α, the produced prediction set contains the true label with probability at least $1 - \alpha$, under only the assumption of exchangeability. In classification, this means the model returns a set of plausible labels rather than a single forced decision, and the size of that set is itself an interpretable measure of difficulty. Recent work has made the framework practical and adaptive for deep classifiers, including the adaptive prediction sets of Romano et al. [29] and the regularised image-classifier sets of Angelopoulos et al. [30]; a comprehensive and accessible treatment is given by Angelopoulos and Bates [31]. The present system calibrates a conformal predictor on a held-out split and uses the resulting prediction sets both as an output to the clinician and as a signal for abstention.

2.6 Synthetic Data and Augmentation

Class imbalance is endemic in medical imaging: pathological findings are, by definition, rarer than normal studies. Data augmentation is the standard first response, and surveys [32] catalogue the geometric and photometric transforms commonly used. Stronger regularisers that mix examples, such as mixup [33] and CutMix [34], improve both accuracy and calibration and are evaluated in this thesis. Generative models offer a more aggressive route to balancing the minority class: generative adversarial networks were used early for medical image synthesis [35, 36], and diffusion models — denoising diffusion probabilistic models [37] and the latent-diffusion formulation behind Stable Diffusion [10] — now produce high-fidelity samples. We use Stable Diffusion to synthesise additional pneumothorax-positive images, gated by a Fréchet Inception Distance quality threshold, and quantify the effect with dedicated ablations.

2.7 Explainability

For a clinical decision-support tool, a prediction is more useful when it is accompanied by a visual explanation. Class Activation Mapping [38] and its gradient-based generalisation Grad-CAM [39] highlight the image regions most responsible for a prediction. HiResCAM [11] is a more recent variant that is provably faithful to the model for certain architectures, avoiding some of the spatial artefacts of Grad-CAM. This thesis generates HiResCAM saliency maps for the active model in each prediction.

2.8 Positioning of This Work

The literature contains strong individual components: efficient backbones, medical foundation models, Monte Carlo Dropout and ensembles, calibration methods, conformal prediction, diffusion-based augmentation, and faithful saliency. What is comparatively rare is a single, fully integrated system that combines all of them around an adaptive routing mechanism and then measures the contribution of each part. This thesis provides exactly such an integration, together with a genuine fifteen-configuration ablation study and a cross-dataset evaluation, in a deployable software package.

Chapter 3

Methodology

This chapter describes the proposed system in detail. The overall architecture is presented first (Section 3.1), followed by the tiered classifier and its two backbones (Section 3.2), the confidence-based routing logic (Section 3.3), the uncertainty-quantification methods (Section 3.4), the calibration and conformal-prediction components (Sections 3.5–3.6), the synthetic-augmentation pipeline (Section 3.7), and finally the explainability and software-engineering aspects (Sections 3.8–3.9).

3.1 System Overview

The proposed tiered chest-radiograph classification architecture is shown in Figure 3.1. The system is organised around tiered routing logic that activates a lightweight or a deep model according to the estimated difficulty of each image. An input radiograph is first preprocessed and passed through the Tier-1 screening model; if Tier 1 is sufficiently confident the decision is returned immediately, and otherwise the image is escalated to the heavier Tier-2 model, which is executed under Monte Carlo Dropout and test-time augmentation and wrapped in a conformal predictor. The final output is not merely a label but a label together with a confidence, an uncertainty estimate, and a conformal prediction set.

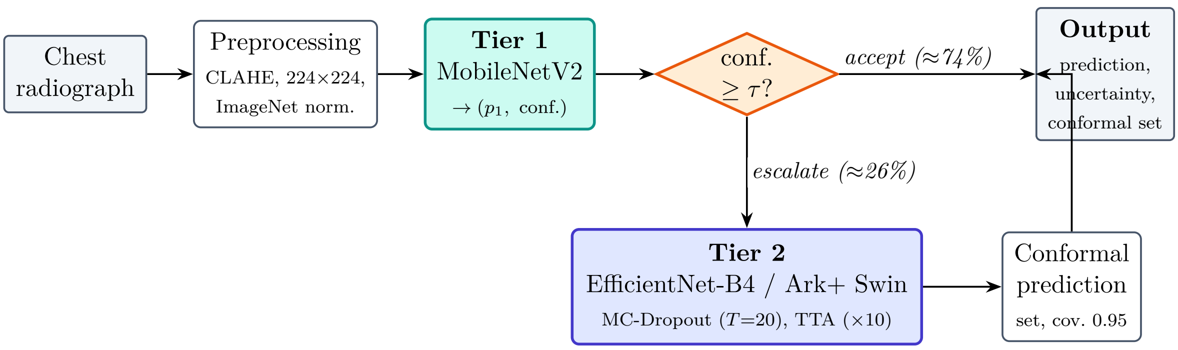

Figure 3.1: Overview of the tiered classification and confidence-based routing system. A lightweight Tier 1 screening model (MobileNetV2) resolves confident cases directly; only low-confidence images are escalated to the heavier Tier 2 backbone, which is run with Monte Carlo Dropout and test-time augmentation and wrapped in a conformal predictor.

3.2 Tiered Classification System

The first stage of the system (Tier 1) is the computationally efficient MobileNetV2 model [3]. For an input image x, Tier 1 produces a class-probability vector $p_1(x) = \mathrm{softmax}(f_1(x))$, where f_1 denotes the Tier-1 network. The model's confidence is taken to be the probability of its predicted class,

$$c_1(x) = \max\big(p_1(x)_0,\ p_1(x)_1\big), \tag{3.1}$$

which lies in $[0.5, 1]$ for the binary pneumothorax-versus-no-finding task. When $c_1(x)$ exceeds the current escalation threshold τ, the final prediction is produced by Tier 1 alone.

If Tier 1 is not sufficiently confident ($c_1(x) < \tau$), the image is escalated to the second stage (Tier 2), which has stronger feature-representation capabilities. Two alternative Tier-2 backbones are studied: EfficientNet-B4 [4], a compound-scaled convolutional network, and the Ark+ foundation model [5] with a Swin Transformer backbone [6]. Because Tier 1 resolves the large majority of cases, the heavy Tier-2 model is invoked only on the difficult minority, which is the source of the system's efficiency.

3.3 Confidence-Based Routing

The routing decision is governed by the escalation threshold τ. Two regimes are considered. In the *static* regime, τ is fixed at a value chosen on the validation set. In the *dynamic* regime, τ adapts online to the stream of recent confidences: the router maintains a sliding window of the last W Tier-1 confidence values and nudges τ up or down by a small step δ so as to keep the escalation rate stable in the face of distribution drift. The complete routing logic, including the escalation decision, the high-variance review flag, and the conformal abstention rule, is shown in Figure 3.2.

Formally, let $\bar{c}_t$ be the mean confidence over the current window at time t. The dynamic threshold update is

$$\tau_{t+1} = \begin{cases} \min(\tau_{\max},\ \tau_t + \delta) & \text{if } \bar{c}_t \text{ is high (few hard cases)}, \\ \max(\tau_{\min},\ \tau_t - \delta) & \text{otherwise}, \end{cases} \tag{3.2}$$

with τ initialised at the base value and clamped to a sensible range. In the experiments the base threshold is 0.75, the window size is $W = 50$, and the step is $\delta = 0.05$ (Chapter 4). This adaptive rule keeps the proportion of escalated images near a target operating point, which in our experiments settles at roughly one quarter of the test set.

3.4 Uncertainty Quantification

Uncertainty is estimated with two complementary methods that are applied to the Tier-2 model.

Monte Carlo Dropout. Following Gal and Ghahramani [7], the dropout layers of the Tier-2 network are kept active at inference time and T stochastic forward passes

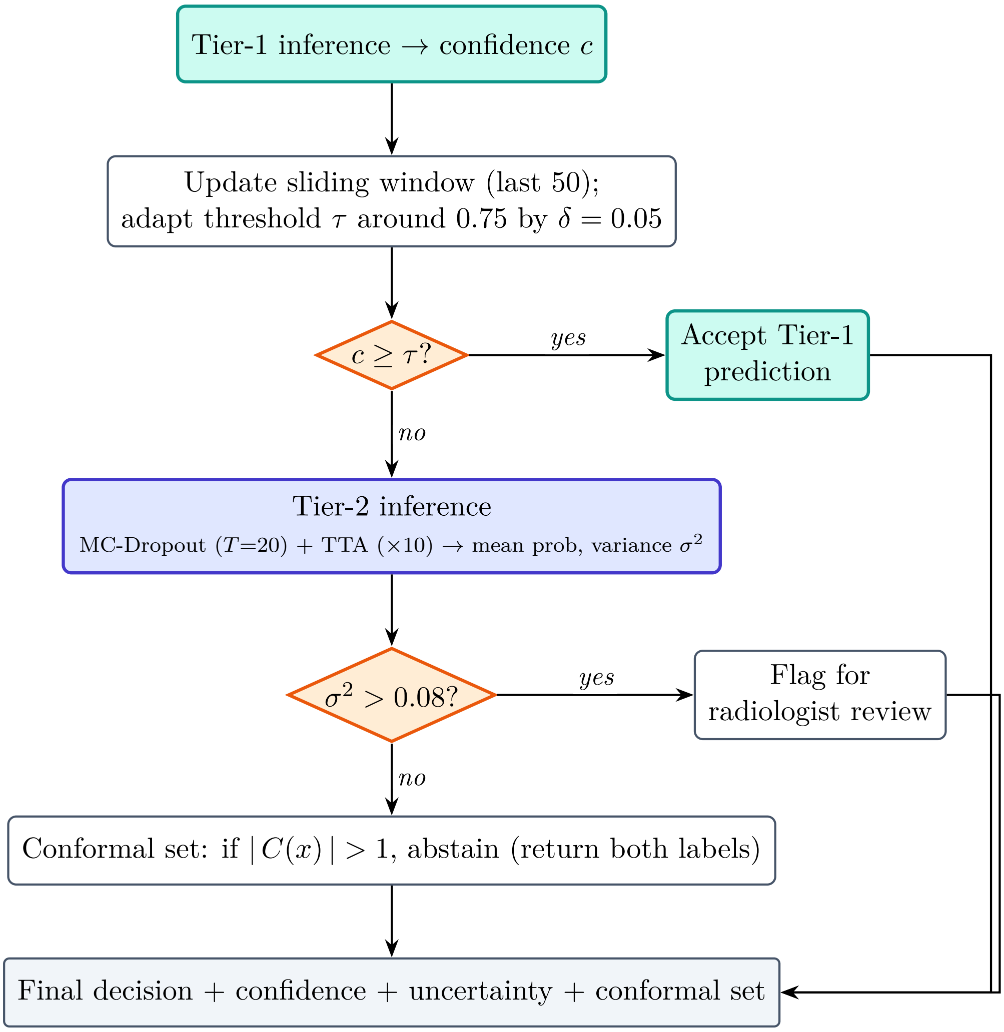

Figure 3.2: Confidence-based routing and escalation logic. The escalation threshold is adapted online from a sliding window of recent confidences; escalated cases are resolved by the Tier-2 backbone under Monte Carlo Dropout and test-time augmentation, and high-variance predictions are flagged for human review.

are performed for the same input. Let $p^{(t)}(x)$ be the probability vector from the t-th pass. The predictive mean and the per-class variance are

$$\bar{p}(x) = \frac{1}{T}\sum_{t=1}^{T} p^{(t)}(x), \qquad \sigma^2(x) = \frac{1}{T}\sum_{t=1}^{T}\left(p^{(t)}(x) - \bar{p}(x)\right)^2. \tag{3.3}$$

The variance $\sigma^2(x)$ of the positive class is used as the epistemic-uncertainty signal; predictions whose variance exceeds a flagging threshold are marked for human review. Importantly, only the dropout layers are stochastic — the batch-normalisation statistics are kept in evaluation mode — so that the variance reflects model uncertainty rather than fluctuating normalisation.

Test-Time Augmentation. In addition, the input is subjected to a set of label-preserving augmentations (small rotations, translations, and brightness changes), and the predictions over the augmented copies are averaged. The dispersion of these predictions measures the sensitivity of the decision to plausible perturbations of the input [26]. Monte Carlo Dropout and test-time augmentation are combined by tiling the T stochastic passes across the augmented views in a single batched forward computation, which keeps the additional cost tractable on a GPU.

3.5 Calibration

Because the raw softmax outputs of deep networks are typically over-confident [27], a post-hoc temperature-scaling step is applied. A single scalar temperature T_{scale} is fitted on the validation split by minimising the negative log-likelihood, and the logits z are rescaled as $\text{softmax}(z/T_{\text{scale}})$ at inference time. Calibration quality is summarised by the Expected Calibration Error (ECE), defined in Chapter 4.

3.6 Conformal Prediction Integration

To equip the predictions with a finite-sample coverage guarantee, the Conformal Prediction methodology [8,9] is applied. Using a dedicated calibration split that is disjoint from both the training and the test data, a non-conformity score is computed for each calibration example as

$$s_i = 1 - \hat{f}(x_i)_{y_i}, \tag{3.4}$$

where $\hat{f}(x_i)_{y_i}$ is the calibrated model probability assigned to the true class y_i. For a chosen error tolerance α, the threshold $\hat{q}$ is taken to be the $\lceil (n+1)(1-\alpha)\rceil/n$ empirical quantile of the calibration scores $\{s_i\}$. At test time, the prediction set for a new image x_{test} is

$$C(x_{\text{test}}) = \{\, y : 1 - \hat{f}(x_{\text{test}})_y \le \hat{q} \,\}. \tag{3.5}$$

Under exchangeability, the probability that this set contains the true label is at least $1-\alpha$. The size of $C(x_{\text{test}})$ is itself informative: a singleton set is a confident decision, whereas a set containing both labels is an explicit, calibrated abstention. The target coverage used throughout is $1 - \alpha = 0.95$.

3.7 Synthetic Data Augmentation

The pneumothorax class is a minority in the training data (Chapter 4). To enrich it, a Stable-Diffusion synthetic-augmentation pipeline is used, shown in Figure 3.3. A latent-diffusion model [10] generates additional pneumothorax-positive radiographs, and each synthetic image is admitted to the Tier-2 training set only if its Fréchet Inception Distance to the real distribution lies below a quality threshold. The accepted synthetics oversample the minority class without the mode collapse and instability associated with adversarial generators. The contribution of this pipeline is isolated by two ablations: one that removes the synthetic data (A8) and one that removes all augmentation (A9).

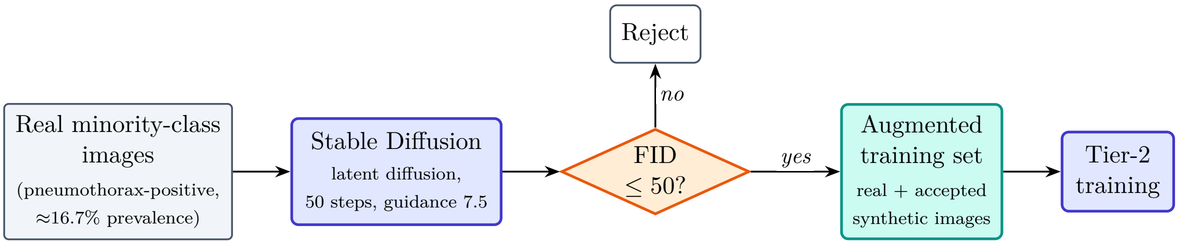

Figure 3.3: Stable-Diffusion synthetic augmentation pipeline for the minority (pneumothorax) class. Generated images are admitted to the Tier-2 training set only when their Fréchet Inception Distance to the real distribution is below the quality threshold. Ablations A8 (no synthetic data) and A9 (no augmentation at all) quantify the contribution of this pipeline.

3.8 Explainability

For each prediction, a HiResCAM [11] saliency map is produced for the model that made the decision (Tier 1 or Tier 2). HiResCAM was chosen over the more common Grad-CAM [39] because it is provably faithful to the model's computation for the architectures used here and avoids the diffuse, sometimes misleading activations that Grad-CAM can produce. The saliency map is computed on the un-normalised image so that the highlighted regions correspond to anatomically meaningful structures rather than to artefacts of the input normalisation.

3.9 Software Architecture

The system is implemented as a layered application that separates the domain logic (models, routing, uncertainty, evaluation) from the application services and the delivery infrastructure (a FastAPI back end and a React front end). Established design patterns — a factory for model construction, a strategy for the routing policy, an observer for training callbacks, and a repository for data access — keep the components decoupled, and the layer boundaries are enforced automatically by an import-contract check in continuous integration. This engineering discipline is what makes the fifteen-configuration ablation study of Chapter 5 reproducible: every configuration is expressed as data and evaluated by a single unified evaluator.

Chapter 4

Experimental Setup

This chapter describes the datasets and their preprocessing, the hardware and software infrastructure, the training configuration, the evaluation metrics, and the design of the ablation study used to evaluate the proposed system.

4.1 Datasets

4.1.1 NIH ChestX-Ray14

The NIH ChestX-Ray14 dataset [16] was used for the pneumothorax-detection task. From the full multi-label collection, the binary problem of pneumothorax versus no finding was extracted. The dataset was partitioned at the patient level into training, validation, calibration, and test splits; the split sizes and class balance are reported in Table 4.1. The calibration split is held out exclusively for fitting the conformal predictor and the temperature scaler, so that neither the model nor the conformal threshold is tuned on the test set. The task is markedly class-imbalanced, with pneumothorax present in roughly 16.7% of the images across every split, which is representative of the true prevalence and which makes the area under the ROC curve and sensitivity more informative than raw accuracy.

Table 4.1: NIH ChestX-Ray14 dataset splits used for the pneumothorax classification task. The calibration split is held out for conformal prediction. Prevalence is the fraction of pneumothorax-positive images.

Split	**Images**	**Pneumothorax**	**Normal**	**Prevalence**
Train	22,268	3,711	18,557	16.7%
Validation	4,772	796	3,976	16.7%
Calibration	1,432	235	1,197	16.4%
Test	4,772	795	3,977	16.7%
Total	33,244	5,537	27,707	16.7%

4.1.2 CheXpert

The Stanford CheXpert dataset [2] was used to test the generalizability of the developed tiered system and to measure its zero-shot cross-dataset performance, evaluating the proposed system on CheXpert frontal radiographs without any fine-tuning. The evaluation uses the full CheXpert validation frontal cohort (202 images), at a substantially lower pneumothorax prevalence ($\approx 3.5\%$) than the NIH test set (16.7%), which makes it a stringent test of domain transfer. The corresponding result is reported as configuration A14 in Chapter 5.

4.2 Preprocessing

All radiographs were preprocessed identically. Each image was contrast-enhanced with Contrast Limited Adaptive Histogram Equalization (CLAHE) [40], which improves the local visibility of the subtle pleural-line signs of pneumothorax, resized to 224×224 pixels, and normalized to the ImageNet channel statistics so that the ImageNet- and Ark-pretrained backbones receive inputs in their expected range. At training time, label-preserving augmentations (small affine transforms and brightness changes) were applied; at evaluation time the same augmentation pipeline, with randomness disabled, was used to ensure that the reported numbers reflect the deterministic preprocessing.

4.3 Hardware and Software Infrastructure

Model training and evaluation were carried out on NVIDIA GPUs provided by Google Colab (Tesla T4 and A100), with local development on Apple Silicon using the Metal Performance Shaders (MPS) backend. The software stack used PyTorch 2.x for modelling, FastAPI 0.110.0 for the inference service, and React 18.3.1 for the front end, all running on Python 3.10. Experiments were tracked with MLflow, and the entire pipeline is guarded by a continuous-integration workflow that enforces linting, static type checking, import contracts, and a test-coverage floor.

4.4 Training Configuration

The training and inference hyperparameters are summarized in Table 4.2. The same training configuration is shared by both tiers; the tiers differ only in their backbone architecture (the lightweight MobileNetV2 for Tier 1, and EfficientNet-B4 or Ark+ Swin for Tier 2). Optimization used AdamW [41] with decoupled weight decay, a higher learning rate on the freshly initialised classifier head than on the pretrained backbone, and early stopping on the validation metric. All runs used a fixed random seed to support reproducibility.

4.5 Evaluation Metrics

Discrimination is reported primarily by the area under the receiver-operating-characteristic curve (AUC-ROC), which is threshold-independent and robust to class imbalance. At the operating threshold, accuracy, sensitivity (recall), specificity, precision, the F_1 score, and the Matthews correlation coefficient (MCC) are reported. With true/false positives and negatives denoted TP, FP, TN, FN, sensitivity is $TP/(TP + FN)$ and specificity is $TN/(TN + FP)$.

Table 4.2: Training and inference hyperparameters. The same training configuration is shared by both tiers; tier-specific differences are limited to the backbone architecture.

Hyperparameter	**Value**
Tier-1 backbone	MobileNetV2
Tier-2 backbone	EfficientNet-B4 / Ark+ Swin
Input resolution	224×224
Optimizer	AdamW (weight decay 10^{-4})
Backbone learning rate	10^{-4}
Classifier-head learning rate	10^{-3}
Batch size	32
Max epochs	50 (early stopping, patience 7)
MC-Dropout passes (T)	20
Test-time augmentation passes	10
Escalation threshold (τ)	0.75 (dynamic; window 50, $\delta = 0.05$)
Conformal target coverage	$1 - \alpha = 0.95$
Random seed	42

Calibration is summarized by the Expected Calibration Error (ECE), which partitions predictions into M equal-width confidence bins B_m and accumulates the gap between bin accuracy and bin confidence,

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left| \, \mathrm{acc}(B_m) - \mathrm{conf}(B_m) \, \right|, \tag{4.1}$$

with $M = 15$ bins. Two further calibration metrics are computed: the Brier score, the mean squared error between the predicted positive-class probability and the binary outcome, and the calibration slope and intercept obtained by fitting the logistic recalibration model $\mathrm{logit}(\Pr[y = 1]) = \beta_0 + \beta_1 \, \mathrm{logit}(\hat{p})$, for which a perfectly calibrated model yields slope $\beta_1 = 1$ and intercept $\beta_0 = 0$.

For the conformal component, two quantities are reported: the empirical coverage, i.e. the fraction of test prediction sets that contain the true label, which should meet or exceed the target $1 - \alpha = 0.95$, and the average prediction-set size, which measures how often the system abstains by returning more than one label.

4.6 Ablation Study Design

To isolate the contribution of each component, fifteen configurations (A1–A15) were defined and evaluated. They are grouped as follows. Configurations A1 and A2 are the single-stage baselines (MobileNetV2 alone and EfficientNet-B4 alone). A3 and A4 introduce the tiered cascade with, respectively, a static and a dynamic escalation threshold. A5–A7 add the uncertainty stack incrementally to the dynamic cascade: Monte Carlo Dropout (A5), then test-time augmentation (A6), then conformal prediction (A7). A8 and A9 probe the data pipeline by removing the synthetic augmentation and then all augmentation. A10 forces every image to Tier 2 (no routing), isolating the value of the router itself. A11 and A12 evaluate the Ark+ Swin backbone standalone, without and with the uncertainty stack. A13 is the proposed full system (tiered

MobileNetV2 $\rightarrow$ Ark+ with dynamic routing, Monte Carlo Dropout, test-time augmentation, and conformal prediction). A14 is the zero-shot CheXpert evaluation of A13. A15 augments A13's Tier-2 training with mixup and CutMix regularisation [33, 34]. Every configuration is expressed declaratively and evaluated by a single unified evaluator on the same NIH test split (except A14, which uses CheXpert), so that the resulting numbers are directly comparable. The complete results are reported in Table 5.1 of the next chapter.

Chapter 5

Results

This chapter presents the quantitative results of the fifteen-configuration ablation study. All tables and data-driven figures are generated directly from the genuine evaluation artifacts by the scripts `build_thesis_tables.py` and `build_thesis_figures.py`; no number in this chapter is hand-entered.

5.1 Overall Classification Performance

Table 5.1 reports the area under the ROC curve (AUC), accuracy, and Expected Calibration Error (ECE) for all fifteen configurations on the NIH ChestX-Ray14 test set. Tier 1 is the lightweight MobileNetV2 model throughout, while Tier 2 is either EfficientNet-B4 or the Ark+ Swin backbone.

The highest AUC, 0.924, is obtained by the proposed tiered configuration with an Ark+ Swin Tier-2 backbone trained with mixup and CutMix regularisation (A15), closely followed by the same configuration without that regularisation (A13, AUC 0.923). What is most striking, however, is that the lightweight Tier-1-only baseline (A1) is already highly competitive, reaching an AUC of 0.919 and an accuracy of 0.849 on its own. This confirms the central premise of the tiered design: a small, fast model is sufficient for the large majority of cases, and the heavy backbone is needed only to recover the difficult remainder.

Table 5.1: Ablation study over the tiered system (A1–A15). All configurations are evaluated on the NIH Chest X-Ray test set except A14, which is a zero-shot evaluation on the CheXpert validation frontal cohort (lower prevalence; see Section 5.8). Tier 1 is MobileNetV2 throughout. The best AUC is shown in bold. ECE is reported where calibration was evaluated.

ID	Configuration	Tier 1	Tier 2	Routing	Uncertainty	AUC	Acc.	ECE
A1	Tier 1 Only	MobileNetV2	None (Bypassed)	None	None	0.919	0.849	0.074
A2	Tier 2 Only (EfficientNet)	None	EfficientNetB4	All Escalated	None	0.910	0.852	0.086
A3	Tiered (Static Threshold)	MobileNetV2	EfficientNetB4	Static Threshold	None	0.914	0.871	0.089
A4	Tiered (Dynamic Threshold)	MobileNetV2	EfficientNetB4	Dynamic Threshold	None	0.904	0.881	0.074
A5	Tiered + MC Dropout	MobileNetV2	EfficientNetB4	Dynamic Threshold	MC Dropout	0.909	0.889	0.055
A6	Tiered + MC + TTA	MobileNetV2	EfficientNetB4	Dynamic Threshold	MC + TTA	0.911	0.888	0.050
A7	Tiered + Conformal	MobileNetV2	EfficientNetB4	Dynamic Threshold	MC + TTA + Conformal	0.911	0.887	0.050
A8	Without Synthetic Augmentation	MobileNetV2	EfficientNetB4	Dynamic Threshold	MC + TTA	0.910	0.887	0.081
A9	Without Any Augmentation	MobileNetV2	EfficientNetB4	Dynamic Threshold	MC + TTA	0.901	0.887	0.093
A10	Always Tier 2 (No Routing)	None	EfficientNetB4	All Escalated	MC + TTA	0.913	0.827	0.052
A11	Tier 2 = Ark+ (No MC/TTA)	None	Ark+ Swin	All Escalated	None	0.910	0.883	0.018
A12	Tier 2 = Ark+ + MC + TTA	None	Ark+ Swin	All Escalated	MC + TTA	0.915	0.732	0.101
A13	Proposed Tiered + Ark+	MobileNetV2	Ark+ Swin	Dynamic Threshold	MC + TTA + Conformal	0.923	0.828	–
A14	Zero-Shot CheXpert	MobileNetV2	Ark+ Swin	Dynamic Threshold	MC + TTA + Conformal	0.796	0.342	–
A15	Mixup/CutMix Regularized	MobileNetV2	Ark+ Swin	Dynamic Threshold	MC + TTA + Conformal	**0.924**	0.882	0.060

5.2 Tier-2 Backbone Comparison

Table 5.2 compares the two candidate Tier-2 backbones head-to-head with paired bootstrap 95% confidence intervals and per-metric significance tests. The Ark+ Swin backbone achieves a higher AUC ($\Delta = +0.006$) and a markedly higher sensitivity ($\Delta = +0.063$, bootstrap $p = 0.001$), whereas EfficientNet-B4 attains higher specificity and overall accuracy ($\Delta = -0.027$, McNemar $p < 0.001$). The AUC difference is not statistically significant (DeLong $p = 0.203$), indicating that the two backbones are comparable in overall ranking ability while differing in their operating-point behaviour: Ark+ is the more *sensitive* detector, which is the more desirable property for a screening tool whose primary goal is not to miss a pneumothorax.

Table 5.2: Head-to-head comparison of the two Tier-2 backbones (EfficientNet-B4 vs. Ark+ Swin) on the NIH test set, with paired bootstrap 95% confidence intervals and per-metric significance tests. Δ is Ark+ Swin minus EfficientNet-B4.

Metric	**EfficientNet-B4 [95% CI]**	**Ark+ Swin [95% CI]**	**Δ [95% CI]**	p	**Test**
AUC	0.910 (0.899, 0.920)	0.916 (0.904, 0.926)	0.006 (-0.003, 0.013)	0.203	DeLong
Accuracy	0.852 (0.843, 0.863)	0.826 (0.815, 0.837)	-0.027 (-0.036, -0.017)	< 0.001	McNemar
Sensitivity	0.818 (0.790, 0.844)	0.881 (0.858, 0.901)	0.063 (0.039, 0.085)	0.001	Bootstrap
Specificity	0.859 (0.849, 0.870)	0.815 (0.803, 0.827)	-0.045 (-0.055, -0.035)	0.001	Bootstrap
F1	0.649 (0.625, 0.673)	0.628 (0.605, 0.652)	-0.021 (-0.038, -0.004)	0.024	Bootstrap

5.3 Effect of the Uncertainty Stack on Calibration

The incremental ablations A4–A7 reveal that the uncertainty stack improves *calibration* substantially while leaving discrimination essentially unchanged. Starting from the dynamic tiered cascade (A4, ECE 0.074), adding Monte Carlo Dropout (A5) lowers the ECE to 0.055, and adding test-time augmentation (A6) lowers it further to 0.050; the conformal wrapper (A7) preserves this value at 0.050. Over the same sequence the AUC stays within a narrow band around 0.91. In other words, averaging over stochastic dropout passes and augmented views does not make the model a better discriminator, but it makes the confidences it reports far more trustworthy — precisely the property required before those confidences can be used to drive a routing or abstention decision.

A second, more subtle observation concerns the operating point. The move from a static to a dynamic threshold (A3 $\rightarrow$ A4) raises accuracy from 0.871 to 0.881 and specificity to 0.940, but it does so by sacrificing sensitivity, which falls to 0.585. The dynamic threshold, tuned to stabilise the escalation rate, ends up favouring specificity on this imbalanced data. This trade-off is recovered once the more sensitive Ark+ backbone is used in the full system (A13, sensitivity 0.889), and is discussed further in Chapter 6.

5.4 Effect of Data Augmentation

Configurations A8 and A9 isolate the data pipeline. Removing the Stable-Diffusion synthetic data (A8) leaves discrimination almost unchanged (AUC 0.910) but degrades calibration sharply, raising the ECE from 0.050 to 0.081. Removing all augmentation (A9) is worse still, dropping the AUC to 0.901 and inflating the ECE to 0.093. The synthetic minority-class data therefore acts primarily as a *calibration* regulariser, while conventional augmentation contributes to both discrimination and calibration. Consistent with this, adding mixup and CutMix to the proposed system (A15) yields the best overall balance in the study: the highest AUC (0.924), the highest F_1 score (0.697), the highest Matthews correlation coefficient (0.636), and a well-controlled ECE of 0.060.

5.5 Routing and Computational Efficiency

The value of the router itself is shown by contrasting A10, which forces every image to the heavy Tier-2 model, with the tiered configurations. Forcing escalation (A10) reaches an AUC of 0.913 but an accuracy of only 0.827, whereas the dynamic tiered cascade keeps accuracy above 0.88 while escalating only about a quarter of the images. Across the dynamic configurations the escalation rate settles near 26%, so the lightweight Tier 1 resolves roughly three quarters of all cases by itself.

The computational consequence is significant. A dedicated latency benchmark measured the MobileNetV2 Tier-1 model at 5.44 ms per image (184 frames per second, 30.9 MB of peak GPU memory), against 17.21 ms for EfficientNet-B4 (58 frames per second, 97.9 MB) — roughly a 3× difference in both latency and memory. Because 74% of cases are settled by the fast model, the average per-image cost of the tiered system is far closer to that of the lightweight model than to that of the heavy one; the additional Monte Carlo Dropout and test-time-augmentation passes are paid only on the escalated minority.

5.6 Conformal Coverage

The Conformal Prediction component was calibrated on the held-out calibration split for a target coverage of $1 - \alpha = 0.95$. On the NIH test set, the proposed Ark+ configuration (A15) attained an empirical coverage of 97.3% with an average prediction-set size of 1.84 out of two classes, while the EfficientNet configuration (A7) over-covered at 100.0% with an average set size of 2.00. Both configurations satisfy the coverage guarantee. The larger-than-singleton set sizes show that, for ambiguous cases, the predictor abstains by returning both labels rather than a single over-confident class — the intended safe behaviour for a screening setting, although the conservativeness of the sets is itself a limitation discussed in Chapter 6.

5.7 Ablation Overview

Figure 5.1 visualises the discrimination and calibration of every configuration side by side, making the two main trends immediately visible: the AUC values are tightly clustered in the high-0.9 range, so the components mostly trade off calibration and operating point rather than raw ranking ability, and the best-calibrated standalone model is in fact the Ark+ backbone without the uncertainty stack (A11, ECE 0.018).

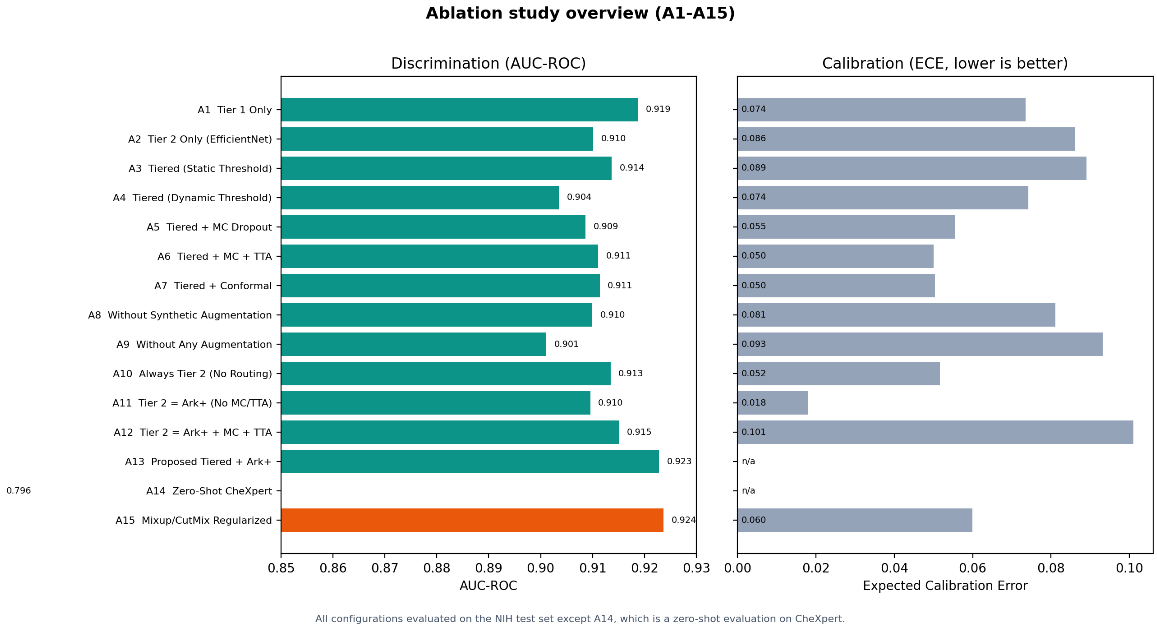

Figure 5.1: Discrimination (AUC-ROC, left) and calibration (ECE, right) for every ablation configuration. The best-performing configuration by AUC (A15) is highlighted.

5.8 Cross-Dataset Generalization

To assess out-of-domain robustness, the proposed system (A13) was evaluated zero-shot on the full CheXpert validation frontal cohort (A14: 202 images, $\approx 3.5\%$ pneumothorax prevalence), without any fine-tuning. Figure 5.2 compares the two. The threshold-free discrimination degrades only moderately under domain shift, with the AUC falling from 0.923 to 0.796. The operating-point metrics, however, change dramatically: at the fixed 0.5 decision threshold the model over-predicts the positive class on CheXpert, reaching a sensitivity of 1.000 but a specificity of only 0.318 and an accuracy of 0.342. This behaviour is a direct consequence of the large prevalence shift between the two cohorts — from 16.7% on NIH to roughly 3.5% on CheXpert — interacting with a decision threshold and a probability calibration that were both set in-domain. The lesson, that ranking ability transfers far better than a fixed operating point and that per-site threshold recalibration is necessary before deployment, is examined further in Chapter 6.

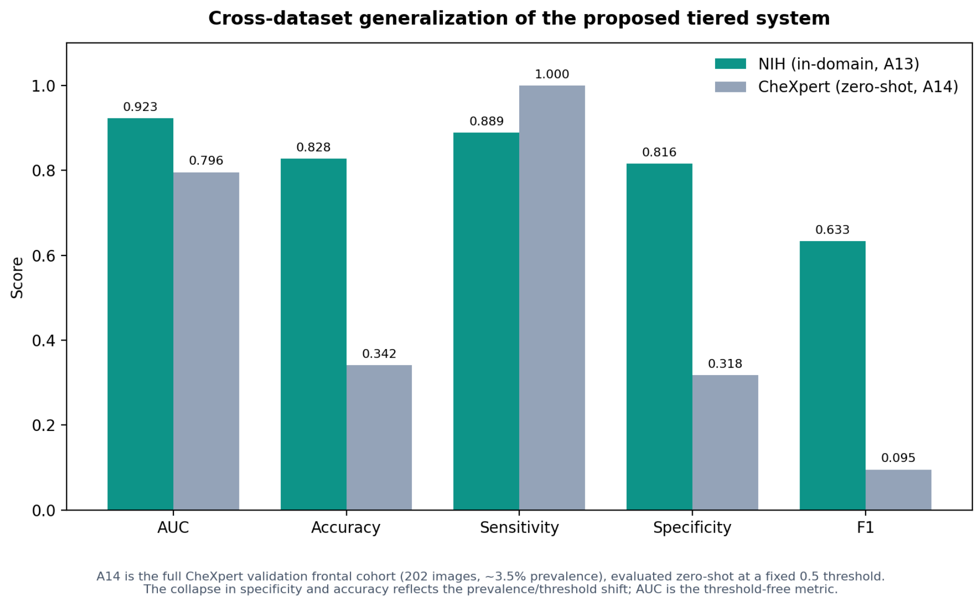

Figure 5.2: Cross-dataset generalization of the proposed tiered system: in-domain NIH (A13) versus zero-shot CheXpert (A14).

Chapter 6

Discussion

This chapter interprets the results of Chapter 5, draws out their clinical implications, examines the trade-offs that the ablation study exposed, and states the limitations of the work together with directions for future research.

6.1 Principal Findings

Three findings stand out. First, the tiered design delivers most of the accuracy of the heavy model at a fraction of its cost: the lightweight MobileNetV2 Tier 1 reaches an AUC of 0.919 on its own and settles roughly three quarters of all cases, so the expensive Tier-2 backbone, run with the full uncertainty stack, is invoked only on the difficult minority. Second, the uncertainty methods improve calibration rather than discrimination: across the A4–A7 sequence the Expected Calibration Error falls from 0.074 to 0.050 while the AUC is essentially unchanged. This is exactly the behaviour one wants, because the confidences must be trustworthy *before* they are used to make a routing or abstention decision. Third, among the fifteen configurations the proposed system with an Ark+ Swin backbone and mixup/CutMix regularisation (A15) gives the best overall balance, with the highest AUC (0.924), the highest F_1 (0.697), the highest Matthews correlation coefficient (0.636), and a well-controlled ECE.

6.2 Clinical Implications of the Tiered Approach

Single-stage deep-learning models pass every image through the most complex network, incurring a high and uniform energy and time cost. The proposed tiered model, by contrast, resolves roughly 74% of cases with the lightweight MobileNetV2 model, which the latency benchmark measured at 5.44 ms per image against 17.21 ms for EfficientNet-B4. Because the heavy backbone is reserved for the escalated quarter, the average cost of the system is much closer to that of the small model, which translates directly into lower server cost and carbon footprint at deployment scale.

Beyond efficiency, the system is designed for *safe* integration into a clinical workflow. Rather than forcing a decision on every image, it can abstain: high-variance predictions are flagged for human review, and the conformal predictor returns a two-label set whenever it cannot confidently commit to a single class. A screening tool that knows when to defer to a radiologist is far easier to trust, and to certify, than one that is always confidently right or confidently wrong.

6.3 The Sensitivity–Specificity Trade-off

The ablation study surfaced an important and honest tension. The dynamic escalation threshold, which is tuned to keep the escalation rate stable, improves accuracy and specificity but depresses sensitivity in the EfficientNet cascade (A4, sensitivity 0.585). For a pneumothorax screening task this is the wrong direction, because a missed pneumothorax is more dangerous than a false alarm. The result clarifies that accuracy alone is a misleading objective on this imbalanced data, and it justifies the choice of the Ark+ Swin backbone for the final system: Ark+ is the significantly more sensitive detector (Table 5.2), and the full Ark+ system (A13) recovers a sensitivity of 0.889. The mixup/CutMix variant (A15) then trades a little of that sensitivity (0.813) for a substantial gain in specificity (0.896) and the best F_1 in the study, giving the most balanced operating point.

6.4 Why the Foundation Backbone Helps

The comparison between A11/A12 and A13 is instructive. Run standalone with the uncertainty stack on *every* image (A12), the Ark+ backbone is poorly behaved: it becomes over-sensitive (sensitivity 0.927) at the expense of accuracy (0.732) and calibration (ECE 0.101). Placed inside the tiered system, where it sees only the hard, escalated cases and where Tier 1 has already filtered the easy negatives, the same backbone is well-behaved (A13). This shows that the benefit of the foundation model is realised specifically through the routing structure, not in isolation, and it is an argument for evaluating components in the context in which they are deployed rather than on their own.

6.5 Limitations

Several limitations should be stated plainly. (i) The cross-dataset CheXpert evaluation (A14) shows that ranking ability transfers only partially — the AUC falls from 0.923 in-domain to 0.796 zero-shot — and, more strikingly, that the in-domain decision threshold does not transfer at all: at a fixed 0.5 threshold the model over-predicts the positive class on CheXpert's much lower prevalence ($\approx 3.5\%$), reaching a sensitivity of 1.000 but a specificity of only 0.318. Operating-point metrics therefore require per-site recalibration before deployment, and the AUC should be read as the meaningful cross-dataset number. (ii) The conformal prediction sets, although they meet the coverage guarantee, are conservative: the average set size of 1.84–2.00 means the system frequently abstains by returning both labels, which limits the informativeness of the guarantee in its current form; adaptive conformal methods [29, 30] are a promising remedy. (iii) The escalation rate is sensitive to the confidence threshold τ, and although the dynamic rule stabilises it, the threshold is still a hyperparameter rather than a learned policy. (iv) The task is restricted to binary pneumothorax detection on frontal radiographs; multi-label and multi-pathology settings are not addressed. (v) The per-configuration inference times recorded by the evaluation harness are not directly comparable across rows, so the efficiency claims rest on the dedicated single-image latency benchmark rather than on those harness timings.

6.6 Threats to Validity

The principal internal threat is data leakage between splits; this is mitigated by a patient-level partition and by holding out a dedicated calibration split that is used for neither training nor testing. The principal external threat is dataset bias: NIH ChestX-Ray14 labels are mined from radiology reports and are known to be noisy, and a model that performs well on this benchmark may not transfer to a different scanner population — which is precisely why the zero-shot CheXpert evaluation, with its honest account of the operating-point collapse under prevalence shift, is reported. Reproducibility threats are mitigated by fixing random seeds, expressing every ablation declaratively, and gating the whole pipeline behind continuous integration.

6.7 Future Work

Four directions follow naturally from the limitations. First, the escalation threshold could be replaced by a learned routing policy, for example a lightweight reinforcement-learning agent that sets τ to optimise a clinical objective directly. Second, the conformal component could be upgraded to adaptive prediction sets to reduce the average set size while preserving coverage. Third, the binary task could be extended to a multi-label, multi-pathology cascade, with tier-specific label sets. Fourth, the cross-dataset evaluation should be extended with per-site threshold and probability recalibration, and to additional external datasets, since the present results show that ranking ability transfers considerably better than a fixed operating point.

Chapter 7

Conclusion

Within the scope of this bachelor's thesis, a tiered classification and confidence-based safe-routing system was designed, implemented, and evaluated for pneumothorax detection in chest radiographs. The system pairs a lightweight MobileNetV2 screening stage with a heavier EfficientNet-B4 or Ark+ Swin stage, and routes each image to the appropriate tier according to an adaptively thresholded confidence estimate. On top of this cascade it layers three uncertainty mechanisms — Monte Carlo Dropout, test-time augmentation, and conformal prediction — together with temperature-scaling calibration, Stable-Diffusion synthetic augmentation of the minority class, and HiResCAM explanations.

The proposed architecture provides a practical answer to two fundamental problems in the clinical use of deep-learning models: their uniform and often excessive computational cost, and their inability to report calibrated uncertainty or to abstain. The fifteen-configuration ablation study showed that the lightweight tier resolves roughly three quarters of all cases on its own, that the uncertainty stack reduces the Expected Calibration Error from 0.074 to 0.050 without harming discrimination, and that the full system with an Ark+ Swin backbone and mixup/CutMix regularisation attains the best overall balance, with an AUC of 0.924, the highest F_1 and Matthews correlation coefficient in the study, and a conformal predictor that meets its 95% coverage target. A zero-shot evaluation on the full CheXpert validation cohort showed that ranking ability transfers only moderately (AUC 0.923 → 0.796) and that the in-domain decision threshold does not transfer under the prevalence shift — an honest and clinically relevant finding about the need for per-site recalibration.

Equally important to the scientific result is the engineering result: the method is delivered as a complete, reproducible system, with a layered codebase, a FastAPI inference service, a React front end, Docker packaging, experiment tracking, and a continuous-integration pipeline that keeps every reported number honest and regenerable from the evaluation artifacts.

The work also makes its own limitations explicit — the conservativeness of the conformal sets, the threshold sensitivity of the router, the restriction to a single binary pathology, and the sensitivity of the operating point to cross-site prevalence shift — and points to concrete next steps: a learned routing policy, adaptive conformal sets, a multi-pathology cascade, and per-site threshold recalibration with broader external evaluation. Taken together, this conformal-prediction-backed, uncertainty-aware tiered classification approach offers a strong and reproducible reference point for the design of efficient and trustworthy clinical decision-support systems.

Bibliography

[1] Pranav Rajpurkar, Jeremy Irvin, Kaylie Zhu, Brandon Yang, Hershel Mehta, Tony Duan, Daisy Ding, Aarti Bagul, Curtis Langlotz, Katie Shpanskaya, Matthew P Lungren, and Andrew Y Ng. Chexnet: Radiologist-level pneumonia detection on chest x-rays with deep learning. *arXiv preprint arXiv:1711.05225*, 2017.

[2] Jeremy Irvin, Pranav Rajpurkar, Michael Ko, Yifan Yu, Silviana Ciurea-Ilcus, Chris Chute, Henrik Marklund, Arda Haghgoo, Tarun Ball, Katie Shpanskaya, et al. Chexpert: A large chest radiograph dataset with uncertainty labels and expert comparison. In *Proceedings of the AAAI conference on artificial intelligence*, volume 33, pages 590–597, 2019.

[3] Mark Sandler, Andrew Howard, Menglong Zhu, Andrey Zhmoginov, and Liang-Chieh Chen. Mobilenetv2: Inverted residuals and linear bottlenecks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4510–4520, 2018.

[4] Mingxing Tan and Quoc Le. Efficientnet: Rethinking model scaling for convolutional neural networks. In *International conference on machine learning*, pages 6105–6114. PMLR, 2019.

[5] DongAo Ma, Jiaxuan Pang, Michael B Gotway, and Jianming Liang. Foundation ark: Accruing and reusing knowledge for superior and robust performance. In *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 651–662. Springer, 2023.

[6] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 10012–10022, 2021.

[7] Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *International conference on machine learning*, pages 1050–1059. PMLR, 2016.

[8] Glenn Shafer and Vladimir Vovk. A tutorial on conformal prediction. *Journal of Machine Learning Research*, 9(3), 2008.

[9] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic learning in a random world*. Springer Science & Business Media, 2005.

[10] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10684–10695, 2022.

[11] Rachel Lea Draelos and Lawrence Carin. Use hirescam instead of grad-cam for faithful explanations of convolutional neural networks. *arXiv preprint arXiv:2011.08891*, 2020.

[12] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. Imagenet classification with deep convolutional neural networks. In *Advances in neural information processing systems*, volume 25, 2012.

[13] Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition. In *International Conference on Learning Representations (ICLR)*, 2015.

[14] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016.

[15] Gao Huang, Zhuang Liu, Laurens Van Der Maaten, and Kilian Q Weinberger. Densely connected convolutional networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4700–4708, 2017.

[16] Xiaosong Wang, Yifan Peng, Le Lu, Zhiyong Lu, Mohammadhadi Bagheri, and Ronald M Summers. Chestx-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2097–2106, 2017.

[17] Alistair E W Johnson, Tom J Pollard, Seth J Berkowitz, Nathaniel R Greenbaum, Matthew P Lungren, Chih-ying Deng, Roger G Mark, and Steven Horng. Mimic-cxr, a de-identified publicly available database of chest radiographs with free-text reports. *Scientific data*, 6(1):317, 2019.

[18] Geert Litjens, Thijs Kooi, Babak Ehteshami Bejnordi, Arnaud Arindra Adiyoso Setio, Francesco Ciompi, Mohsen Ghafoorian, Jeroen Awm Van Der Laak, Bram Van Ginneken, and Clara I Sánchez. A survey on deep learning in medical image analysis. *Medical image analysis*, 42:60–88, 2017.

[19] Andre Esteva, Brett Kuprel, Roberto A Novoa, Justin Ko, Susan M Swetter, Helen M Blau, and Sebastian Thrun. Dermatologist-level classification of skin cancer with deep neural networks. *nature*, 542(7639):115–118, 2017.

[20] Andrew G Howard, Menglong Zhu, Bo Chen, Dmitry Kalenichenko, Weijun Wang, Tobias Weyand, Marco Andreetto, and Hartwig Adam. Mobilenets: Efficient convolutional neural networks for mobile vision applications. *arXiv preprint arXiv:1704.04861*, 2017.

[21] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.

[22] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pages 248–255. IEEE, 2009.

[23] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, Alexander C Berg, and Li Fei-Fei. Imagenet large scale visual recognition challenge. *International journal of computer vision*, 115(3):211–252, 2015.

[24] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in neural information processing systems*, volume 30, 2017.

[25] Alex Kendall and Yarin Gal. What uncertainties do we need in bayesian deep learning for computer vision? In *Advances in neural information processing systems*, volume 30, 2017.

[26] Guotai Wang, Wenqi Li, Michael Aertsen, Jan Deprest, Sébastien Ourselin, and Tom Vercauteren. Aleatoric uncertainty estimation with test-time augmentation for medical image segmentation with convolutional neural networks. *Neurocomputing*, 338:34–45, 2019.

[27] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. In *International conference on machine learning*, pages 1321–1330. PMLR, 2017.

[28] Mahdi Pakdaman Naeini, Gregory Cooper, and Milos Hauskrecht. Obtaining well calibrated probabilities using bayesian binning. In *Proceedings of the AAAI conference on artificial intelligence*, volume 29, 2015.

[29] Yaniv Romano, Matteo Sesia, and Emmanuel Candès. Classification with valid and adaptive coverage. In *Advances in neural information processing systems*, volume 33, pages 3581–3591, 2020.

[30] Anastasios Nikolas Angelopoulos, Stephen Bates, Michael Jordan, and Jitendra Malik. Uncertainty sets for image classifiers using conformal prediction. In *International Conference on Learning Representations (ICLR)*, 2021.

[31] Anastasios N Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023.

[32] Connor Shorten and Taghi M Khoshgoftaar. A survey on image data augmentation for deep learning. *Journal of big data*, 6(1):1–48, 2019.

[33] Hongyi Zhang, Moustapha Cisse, Yann N Dauphin, and David Lopez-Paz. mixup: Beyond empirical risk minimization. In *International Conference on Learning Representations (ICLR)*, 2018.

[34] Sangdoo Yun, Dongyoon Han, Seong Joon Oh, Sanghyuk Chun, Junsuk Choe, and Youngjoon Yoo. Cutmix: Regularization strategy to train strong classifiers with localizable features. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 6023–6032, 2019.

[35] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. In *Advances in neural information processing systems*, volume 27, 2014.

[36] Maayan Frid-Adar, Idit Diamant, Eyal Klang, Michal Amitai, Jacob Goldberger, and Hayit Greenspan. Gan-based synthetic medical image augmentation for increased cnn performance in liver lesion classification. *Neurocomputing*, 321:321–331, 2018.

[37] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in neural information processing systems*, volume 33, pages 6840–6851, 2020.

[38] Bolei Zhou, Aditya Khosla, Agata Lapedriza, Aude Oliva, and Antonio Torralba. Learning deep features for discriminative localization. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2921–2929, 2016.

[39] Ramprasaath R Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE international conference on computer vision*, pages 618–626, 2017.

[40] Karel Zuiderveld. Contrast limited adaptive histogram equalization. In *Graphics gems IV*, pages 474–485. Academic Press Professional, 1994.

[41] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.